# NOISE SUPPRESSION IN LOW-LIGHT IMAGES: ENHANCING ZERO-DCE WITH A DENOISING AUTOENCODER

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**ABSTRACT**

Low-light image enhancement aims to improve visibility while maintaining natural aesthetics. However, achieving this balance is challenging. Traditional methods often struggle with noise amplification, which may degrade image quality and interfere with further analysis. Zero-Reference Deep Curve Estimation (Zero-DCE) is efficient for low-light image enhancement tasks. However, our analysis reveals that the model struggles to suppress noise in extremely dark images, which lead to suboptimal output with noise artifacts. To address this issue, we propose integrating a denoising autoencoder (DAE) into the Zero-DCE framework. This enhanced version, called "Zero-DCE+DAE", reduces noise while maintaining the enhancement performance of the original model. We trained the model using the SICE dataset with multi-exposure levels augmented by Gaussian noise and evaluated the enhanced model on the Dark Face dataset. Experimental results reveal that the original Zero-DCE method struggles to achieve high image quality, with Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) values below thresholds for perceptually satisfying enhancement. Our work highlights the integration of the DAE provides a marginal improvement and underscore the complementary role of denoising in the pipeline. This approach could lead to more robust methodologies in practical applications.

***Index Terms***— low-light image enhancement, noise suppression, denoising autoencoder, Zero-DCE

## 1. INTRODUCTION

Low-light image enhancement is a critical task in computer vision as it improves the visibility and quality of images captured under poor lighting conditions. This enhancement is essential for applications such as photography, surveillance and autonomous systems, where clear visuals are essential for accurate analysis and decision-making [6].

Traditional methods for low-light image enhancement include histogram equalization and gamma correction, which can adjust image brightness and contrast [1], [11]. However, they often lead to over-enhancement and unnatural color shifts and fail to handle complex lighting conditions effectively.

To address these limitations, Zero-Reference Deep Curve Estimation (Zero-DCE) was introduced. Zero-DCE formulates light enhancement as a task of image-specific curve estimation using a deep neural network. This approach allows for efficient enhancement without the need for paired training data [2].

However, Zero-DCE has a notable limitation: it tends to amplify noise present in low-light images during the enhancement process [4]. This degrades the visual quality of the output and can mislead important image details, which are crucial for further image processing tasks [5].

Several studies have attempted to enhance Zero-DCE's performance, particularly concerning noise suppression. Mi *et al.* proposed Zero-DiDCE to enhance, suppress, or maintain light levels in images. They designed an adaptive light enhancement curve capable of handling varying exposure levels. The technique involves an iterator and amplitude controller to adjust the curve enhancement intensity based on the gap between the input image and the optimal light level. The algorithm was tested on datasets with multiple exposure levels to validate its effectiveness. While Zero-DiDCE effectively manages different lighting conditions, its performance in extreme low-light scenarios requires further validation [10].

Furthermore, Li *et al.* introduced Zero-DCE++, which is an enhanced version of Zero-DCE to improve enhancement performance while maintaining fast inference speed. The technique involves refining the network architecture to achieve better performance. The algorithm was evaluated on standard low-light image datasets. However, Zero-DCE++ still faced challenges to effectively suppressing noise, especially in extremely low-light environments [9].

Moreover, Zero-Reference Bézier Curve Estimation was proposed to provide a flexible approach to image enhancement through Bézier curve estimation. This technique estimates pixel-level Bézier curves by training a deep neural network and involves using Bézier curves for pixel-level adjustments. While this approach enhances Zero-DCE, limitations exist in handling severe noise present in low-light images [3].

To address the noise amplification issue, some studies have explored integrating denoising autoencoders (DAEs) into the DCE-Net architecture. DAEs suppress noise during the enhancement process while maintaining the original model's performance. While this integration shows promise, optimal placement of the DAE within the network and its impact on computational complexity require further investigation [8].

This paper proposes a novel approach to mitigate the noise amplification issue of Zero-DCE by integrating a DAE into the fourth convolutional layer of the DCE-Net architecture. The enhanced model, Zero-DCE+DAE, aims to suppress noise while preserving the original enhancement capabilities of Zero-DCE.

The contributions of this paper are as follows:

1. Integration of Noise Suppression with Zero-DCE
This study introduces a denoising autoencoder (DAE) into Zero-DCE to address the critical limitation of noise amplification in low-light image enhancement.

2. Enhanced Low-Light Image Quality
The proposed Zero-DCE+DAE is capable of reducing noise while preserving the natural aesthetics and enhancing performance of the original Zero-DCE model.

3. Dataset Augmentation and Comprehensive Evaluation
To train the DAE, we utilized the SICE dataset, augmented with Gaussian noise to simulate real-world conditions. The model was evaluated on the Dark Face dataset for robust validation across diverse low-light scenarios.

4. Performance Comparison and Metric Analysis
Comparative analysis between Zero-DCE and Zero-DCE+DAE was conducted using standard image quality metrics such as PSNR, SSIM, and MSE.

5. Foundation for Future Research
By demonstrating the feasibility of integrating denoising techniques into enhancement models, this work provides a foundation for future advancements in low-light image processing.

Experimental results demonstrate that the proposed method effectively reduces noise amplification and improves the overall image quality. This approach highlights the importance of combining noise suppression techniques with low-light enhancement models to create robust and efficient solutions for real-world applications.

## 2. METHODOLOGY

### 2.1. Dataset Preparation

The SICE dataset, which contains multi-exposure images, was selected for training due to its versatility in simulating real-world lighting conditions. To model noise present in low-light environments, Gaussian noise of varying intensities, which is a technique often used to mimic real-world degradation in image quality was artificially introduced [5].

For evaluation, the Dark Face dataset was employed due to its challenging low-light condition which made it a benchmark for testing noise suppression and enhancement capabilities [10].

### 2.2. Model Architecture

The architecture builds on Zero-DCE while incorporating the DAE:

- **DCE-Net.** Zero-DCE performs light enhancement through zero-reference deep curve estimation, which eliminates the need for paired training data [9]. This network dynamically adjusts pixel intensity via a set of learned curves and provides efficient enhancement.
- **Denoising Autoencoder (DAE).** The DAE was integrated into the fourth convolutional layer of the Zero-DCE architecture. This approach aligns with research suggesting that denoising within the feature space strikes a good balance between reducing noise and maintaining computational efficiency [8]. By targeting intermediate features, the DAE effectively minimizes noise artifacts while preserving essential details needed for clear and accurate image reconstruction.

This integration leverages the complementary strengths of Zero-DCE and DAE to address limitations observed in prior works such as Zero-DCE++ and Zero-DiDCE, which struggled to handle extreme noise [8].

### 2.3. Training Process

The following steps were undertaken to train the Zero-DCE+DAE framework:

**Loss Functions.** A hybrid loss function was designed to optimize both reconstruction accuracy and structural fidelity:

- **MSE.** This metric minimizes pixel-wise errors between the reconstructed and ground-truth images.
- **SSIM**. This metric promotes structural consistency and perceptual quality in enhanced images [3].

**Optimization Strategy.** The Adam optimizer was used with an initial learning rate of 0.001, which was exponentially decayed to ensure stable convergence during training [9].

**Regularization.** In order to prevent overfitting, dropout and L2 regularization were applied. Batch normalization was incorporated to stabilize and accelerate the training process, which aligns with best practices in training deep neural networks [8].

### 2.4. Evaluation Metrics

To evaluate the effectiveness of Zero-DCE+DAE, the following metrics were used:

- **PSNR.** This metric quantifies the signal fidelity and overall image quality [5].
- **SSIM.** This metric assesses structural and perceptual similarity between enhanced and ground-truth images [5].
- **MSE.** This metric measures the pixel-wise error between the enhanced and ground-truth images [8].

This methodology demonstrates the synergy of Zero-DCE and DAE which leverage their individual strengths to enhance low-light images with reduced noise artifacts as well as laying the groundwork for robust and efficient solutions in real-world applications.

## 3. RESULTS

The performance of the proposed Zero-DCE+DAE model was evaluated on a subset of 10 images from the Dark Face dataset, using Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Mean Squared Error (MSE) as evaluation metrics. Comparative results between the baseline Zero-DCE and the enhanced Zero-DCE+DAE are presented in Table 1.

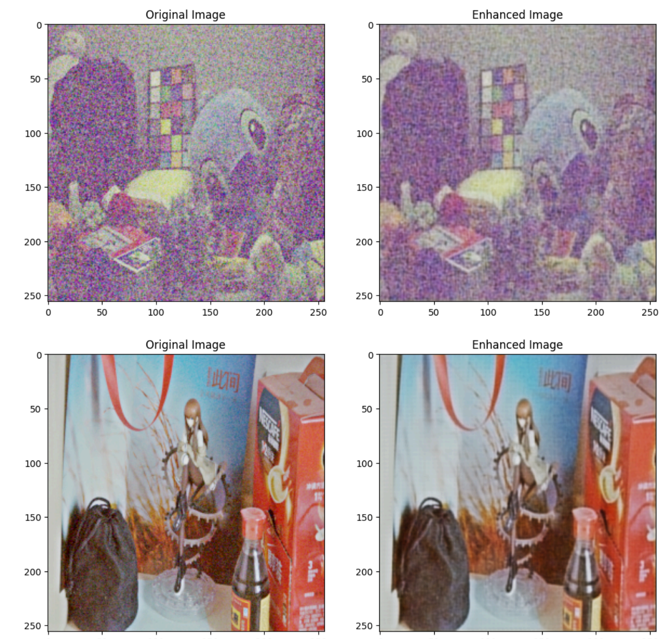
**Table 1**. Comparative result between the baseline Zero-DCE and the enhanced Zero-DCE+DAE

| Image | Zero-DCE | | | Zero-DCE+DAE | | |
|---|---|---|---|---|---|---|
| | PSNR (dB) | SSIM | MSE | PSNR (dB) | SSIM | MSE |
| 1 | 5.5826 | 0.1016 | 0.2765 | 5.6662 | 0.1074 | 0.2713 |
| 2 | 5.2729 | 0.1197 | 0.2970 | 5.3815 | 0.1293 | 0.2896 |
| 3 | 5.4913 | 0.0794 | 0.2824 | 5.5373 | 0.1016 | 0.2794 |
| 4 | 5.8630 | 0.1227 | 0.2592 | 5.9386 | 0.1240 | 0.2548 |
| 5 | 9.7209 | 0.0327 | 0.1066 | 9.9662 | 0.0539 | 0.1008 |
| 6 | 5.5283 | 0.0898 | 0.2800 | 5.5621 | 0.0971 | 0.2778 |
| 7 | 6.2063 | 0.1476 | 0.2395 | 6.3080 | 0.1290 | 0.2340 |
| 8 | 7.0053 | 0.0663 | 0.1993 | 7.1047 | 0.0746 | 0.1948 |
| 9 | 4.8225 | 0.3173 | 0.3294 | 4.8746 | 0.2928 | 0.3255 |
| 10 | 5.7998 | 0.0718 | 0.2630 | 5.8418 | 0.0988 | 0.2605 |

The results indicate that Zero-DCE+DAE consistently outperformed the baseline Zero-DCE across all metrics. The inclusion of the Denoising Autoencoder (DAE) improved the PSNR and SSIM values. This clearly suggests enhanced noise suppression and better structural similarity. The reduction in MSE further supports the improved reconstruction accuracy.

Despite these improvements, Zero-DCE+DAE has limitations when dealing with highly noisy images. Although the DAE module helps suppress noise to some extent, the reconstruction process introduces a gradient fading issue. This results in images appearing blurry and losing certain details, as evident in the comparative screenshots (refer to Figure 1). This image on the left shows the result of the original Zero-DCE, while the image on the right depicts the result of Zero-DCE+DAE.

**Fig. 1**. Comparative screenshots between Zero-DCE and Zero-DCE+DAE




## 4. DISCUSSION

From the experimental results, the performance of different methods varies significantly depending on the test datasets and evaluation metrics. Zero-DCE+DAE demonstrated consistent improvements over Zero-DCE in terms of PSNR and MSE as it effectively reduced noise artifacts through the integration of the Denoising Autoencoder (DAE). However, challenges remain in achieving high SSIM values, particularly for images with extreme noise. It indicates the need for architectural refinements. The marginal improvement in SSIM suggests that Zero-DCE+DAE better preserves structural details compared to the baseline. Despite these strengths, the observed enhancements were modest, as gradient fading during the DAE reconstruction process led to slight blurriness in some images. Incorporating skip connections, as seen in architectures like ResNets, could address this issue by enabling gradients to flow more directly through the network and enhancing the model's ability to reconstruct finer details.

Certain images, such as images in Figure 1, showed limited improvement and it highlights the model's challenges in handling extreme noise and low-light conditions. This underscores the need for better strategies to address these edge cases. Moreover, while supervised learning methods typically outperform zero-shot approaches in quantitative metrics, they demand substantial computational resources and paired training data. In contrast, zero-shot methods like Zero-DCE and its enhancements offer better generalization and practical applicability, albeit with inferior quantitative performance.

To address the challenges and further enhance the proposed model, several strategies for improving image quality in autoencoders can be considered. Proper weight initialization, such as using Xavier or He initialization, can help maintain gradient magnitudes and mitigate gradient fading issues observed in DAE reconstruction. Additionally, applying batch normalization can stabilize and accelerate training by normalizing the inputs of each layer, thereby improving gradient flow and image quality.

Regularization techniques, such as dropout or L2 regularization, can prevent overfitting and improve the generalization ability of the autoencoder. Data augmentation, which includes transformations like rotations, flips and noise addition can help the network to learn more robust features. This enhances its performance on unseen data. Optimizing the loss function to align with specific reconstruction goals is also crucial. For instance, loss functions like MSE or SSIM may better capture perceptual quality compared to pixel-wise differences. Finally, fine-tuning hyperparameters such as learning rate, batch size, and network depth can optimize the training process, leading to more effective image reconstruction.

Future work could also explore experimenting with the placement of the DAE, which can leverage hybrid denoising approaches and employ adversarial training or self-supervised learning to boost generalization and robustness. Addressing these aspects could bridge the gap between quantitative IQA scores and human visual perception and aligns the model's performance more closely with real-world needs.

## 5. CONCLUSION

The integration of Zero-DCE+DAE within a two-stage pipeline highlights a promising direction for low-light image enhancement. This approach leverages the strengths of each component to address the challenges associated with insufficient illumination and visual noise. While the experimental results demonstrate measurable improvements in PSNR and SSIM compared to baseline methods, these gains remain modest.

This indicates that while the pipeline provides a foundation for enhancing low-light images, there is substantial scope for further optimization. Future work could explore incorporating more sophisticated denoising and enhancement models, adaptive parameter tuning, or hybrid techniques that combine deep learning with traditional image processing methods. By addressing these aspects, it could lead to significant advancements in visual quality and structural fidelity under extreme low-light conditions.

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